INDX Filament Console

Setup guide for a Raspberry Pi 5 with the 7" Touch Display 2

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What you need

- [ ] Raspberry Pi 5 (2 GB or more)
- [ ] Official Raspberry Pi 27 W USB-C power supply
- [ ] Raspberry Pi Touch Display 2, 7" (comes with its display and power cables)
- [ ] microSD card, 16 GB or larger
- [ ] USB-A to USB-C **data** cable, Pi to printer (some cables only charge; those won't work)
- [ ] USB keyboard and mouse, for setup only
- [ ] A computer to prepare the microSD card, and a USB stick to carry the console file
- [ ] The console file: `indx-console.zip`

No AI HAT, camera or other add-on board is needed.

Setup

1 Prepare the microSD card

1. On your computer, install and open **Raspberry Pi Imager** (free, from raspberrypi.com/software).
2. Choose **Raspberry Pi 5**, then **Raspberry Pi OS (64-bit)** (the version with the desktop), then your microSD card.
3. When asked to customise settings, set a **username and password**, your **Wi-Fi**, and a **hostname** such as `indx-console`.
4. Write the card, then put it in the Pi.

2 Attach the screen

1. Connect the display's ribbon cable to one of the Pi's **CAM/DISP** ports.
2. Connect the display's power lead to the Pi's GPIO pins, as shown in the leaflet that comes with the display.
3. Screw the Pi onto the back of the display.

3 First start and update

1. Plug in the keyboard, mouse and power supply. The desktop appears in portrait.
2. Open **Terminal** (the black window icon in the top bar) and type:

```
sudo apt update && sudo apt full-upgrade -y
sudo reboot
```

The update can take 5–15 minutes. Enter the password you set in step 1 if asked.

4 Copy the console to the Pi

1. Copy `indx-console.zip` onto a USB stick and plug it into the Pi.
2. In the File Manager, drag `indx-console.zip` into your **Home** folder.

5 Install

Open Terminal and type these lines, pressing Enter after each:

```
cd ~
unzip -o indx-console.zip
cd indx-console
bash install.sh
sudo reboot
```

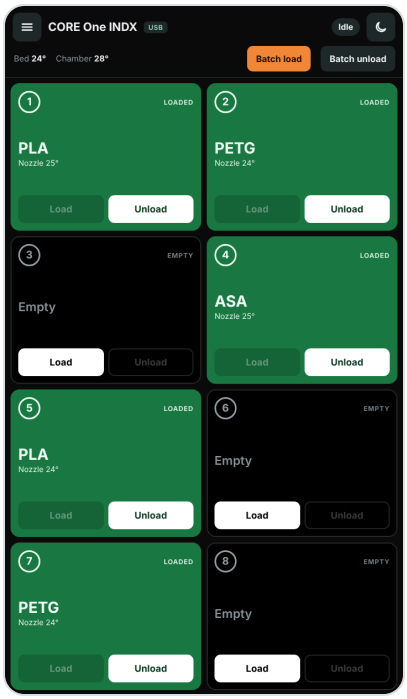
The install takes 2–5 minutes. After the reboot, the console opens full-screen by itself.

6 Connect the printer

1. Plug the USB-A to USB-C cable into any USB port on the Pi and into the printer's USB-C port.
2. Make sure the printer is on and its firmware is up to date.

The Pi's own USB-C port is for power only. Keep using the Pi's power supply.

7 Check that it works



- A **USB** badge appears next to the printer name.
- Within about 30 seconds every tool shows its material, or **Empty**.
- Bed and chamber temperatures appear at the top.

Done. The steps below are optional extras.

If the USB badge never appears, try another cable (it must carry data, not just charge), then see Troubleshooting.

8 Wi-Fi backup OPTIONAL

Keeps the console working if the USB cable comes loose.

1. Tap the menu button (☰) in the top left, then **Wi-Fi backup**.
2. Tap **Get address from printer**.
3. Enter the **PrusaLink password**, found on the printer under **Settings → Network → PrusaLink**.
4. Tap **Test connection**, then **Save**.

Loading over Wi-Fi needs a USB flash drive plugged into the printer. Typing is easier from a phone: open `http://indx-console.local:8080` (use your hostname from step 1).

9 Side filament sensors OPTIONAL

Lets a load start by itself when you feed filament, and puts tools that already have filament first in line.

1. Menu (☰) → **Filament sensors** → **Turn on sensor reporting**.
2. Confirm on the printer's screen.

Using the console

WHAT	HOW
Card colours	🟩 Green = loaded ⬛ Black = empty 🟨 Yellow = being set up, loading or unloading
Load one tool	Tap **Load** on an empty card, pick the material, then **Load now** or **Add to load list**. Follow the printer's prompts.
Unload	Tap **Unload** on a green card and confirm.
Several tools	**Batch load** or **Batch unload** (top right). The big number on each yellow card is its place in line.
During a print	The progress screen shows automatically. Tools are locked until the print ends. The X shows the tool cards.
Spool details	Tap a card to add brand, colour and weight, which tracks the amount left.
Screen off	Moon button, top right. Tap anywhere to wake. The screen also sleeps on its own (set in the menu).
Brightness, sleep	Menu (☰).
Leave the console	Menu (☰) → **Exit to desktop**. Reopen it with the **INDX Console** icon on the desktop.

Updating

Copy the new `indx-console.zip` to the Home folder (step 4) and run step 5 again. Your settings and spool details are kept.

Troubleshooting

PROBLEM	TRY THIS
No USB badge	Use a data cable, re-plug both ends, and check that the printer is on. Menu → **Show printer log** shows what the printer is sending.
Screen is black	It's asleep. Tap it once; that tap doesn't press anything.
Console didn't open at start-up	Double-tap the **INDX Console** desktop icon. If it keeps happening, run step 5 again.
Asked for a "keyring" password	Run step 5 again with the latest `indx-console.zip`.
Something else	In Terminal, `journalctl -u indx-console -f` shows the console's log.